

Project 2*

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1 Introduction

the motivation for the analysis

Can college readiness, staff demographics, and school demographics predict graduation rates in California high schools?

a brief overview of the broader context

the knowledge gap that is addressed by the paper

what was done in the analysis

the key findings and why they are important

the structure of the paper

This project draws from the data sources Common Core of Data, Small Area Income and Poverty Estimates, The Civil Rights Data Collection, *EDFacts*, Integrated Postsecondary Education Data System, College Scorecard, National Historical Geographic Information System, Federal Student Aid, National Association of College and University Business Officers, National Center for Charitable Statistics, Model Estimates of Poverty in Schools (MEPS), Equity in Athletics Data (EADA), and Campus Safety and Security (2025a).

*Project repository available at: https://github.com/meganajoseph/261a_project2.

2 Data

The data are from the Education Data Explorer from the Urban Institute's Center on Education Data and Policy. This tool compiles data from national data sources on schools, districts, and colleges in order to assist in creating insights to improve student outcomes (2025a). I used this [query](#) to produce a dataset of high schools in California from 2016-2017 with information on school characteristics, absenteeism, college readiness, course offerings, student demographics, discipline, school safety, student outcomes, and teachers and staff (Education Data Portal (Version 0.23.0), Urban Institute 2023).

[talk more about NA values, and transformations]

The dataset has 2125 observations while the original has 6111 observations.

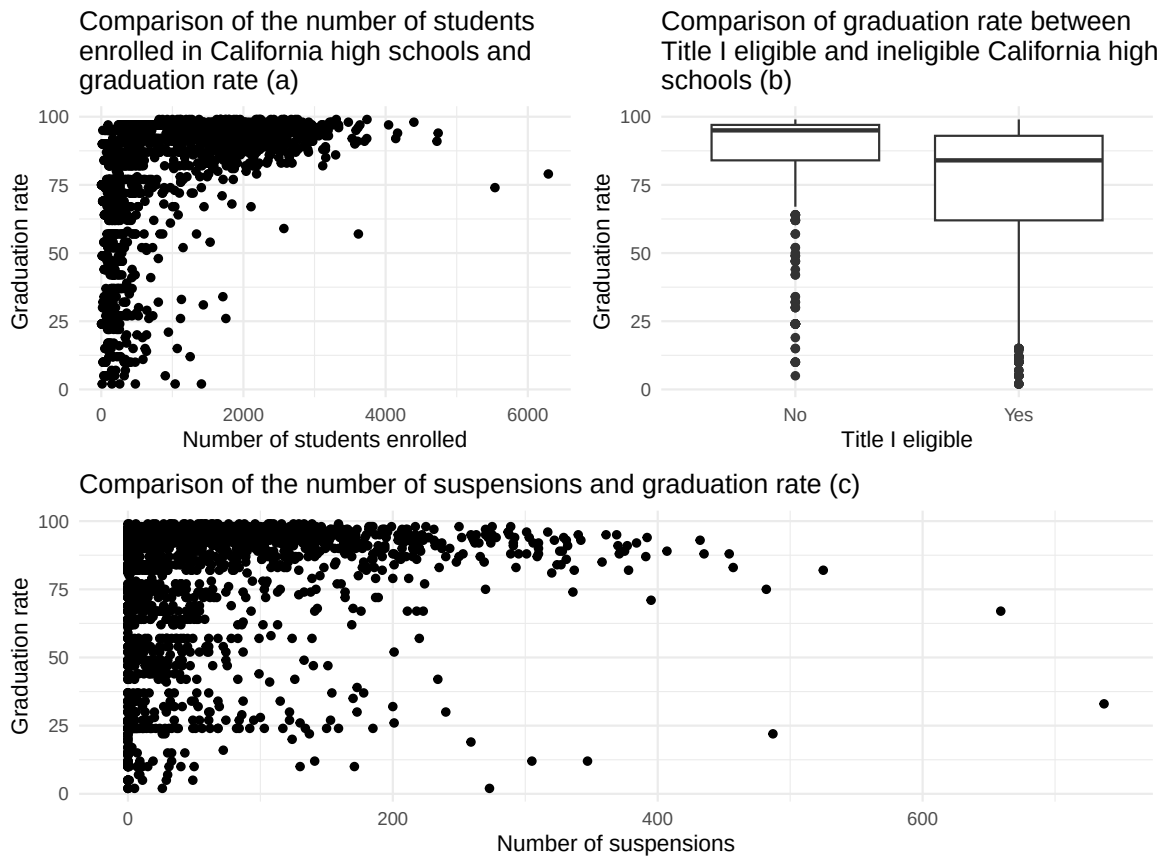


Figure 1: Plot (a) shows a scatter plot of the graduation rate over the number of students enrolled. Plot (b) shows a boxplot of graduation rate at Title I eligible and ineligible schools. Plot (c) shows the relationship between the number of suspensions in a school (x-axis) and graduation rate (y-axis).

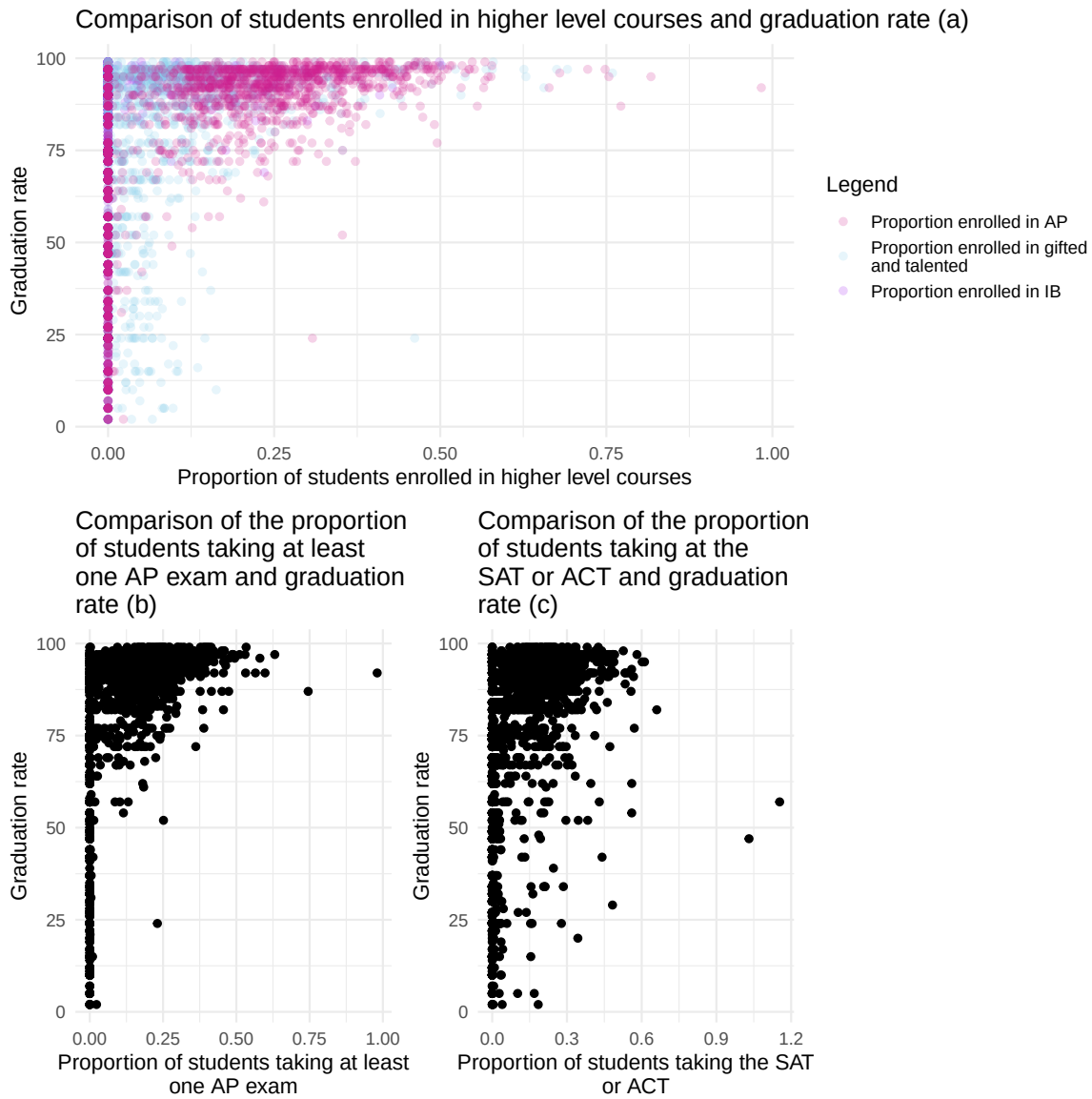


Figure 2: Plot (a) shows a scatter plot showing the relationship between the proportion of students enrolled in high level courses including the IB program (purple), AP program (pink), and gifted and talented programs (blue) (x-axis) and graduation rate (y-axis). Plot (b) shows a scatter plot of the graduation rate over the proportion of students taking at least one AP exam. Plot (c) shows a scatter plot of the graduation rate over the proportion of students taking either the SAT or ACT.

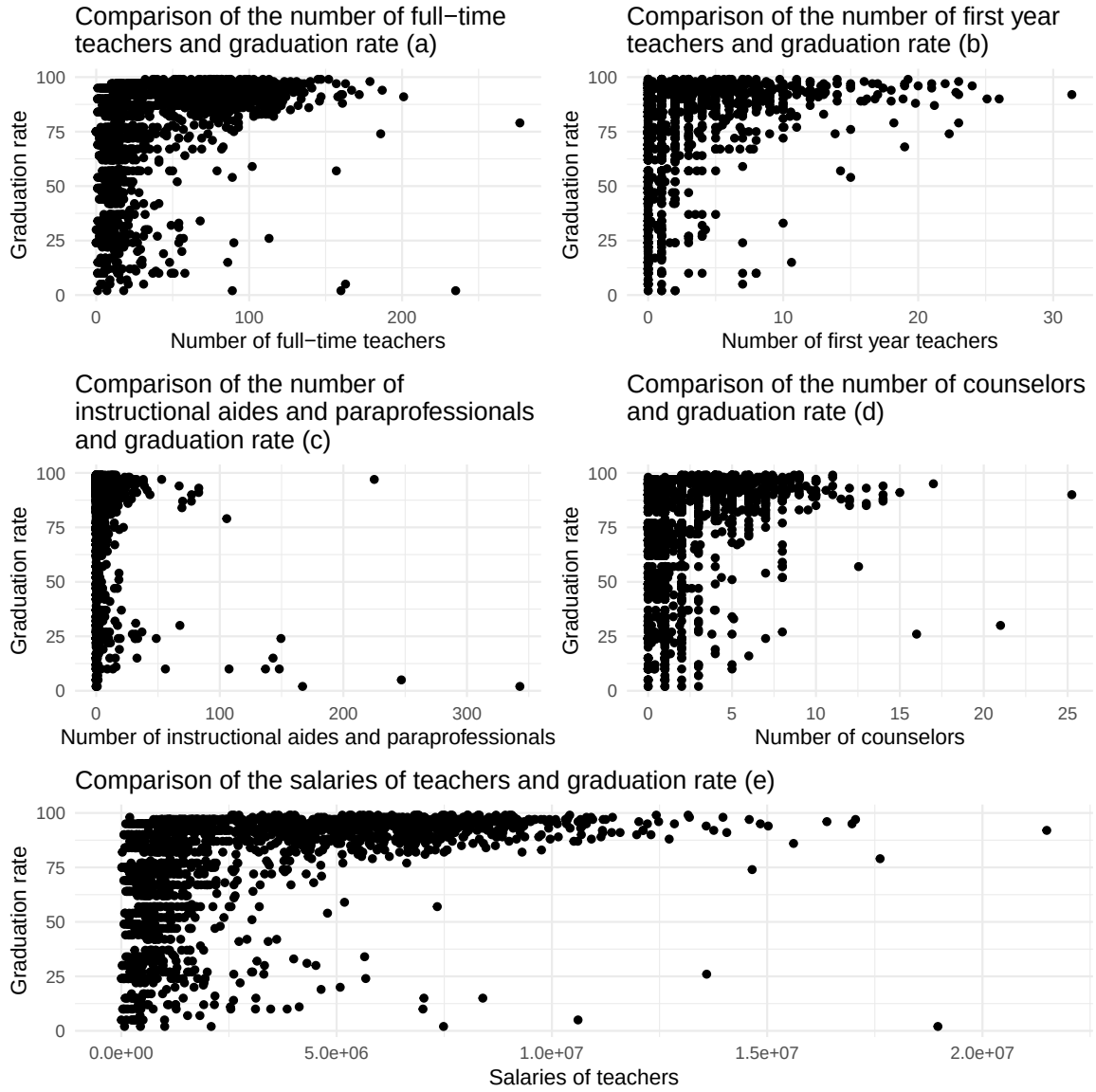


Figure 3: Plot (a) shows the relationship between the number of full-time teachers (x-axis) and graduation rate (y-axis). Plot (b) shows the relationship between the number of first year teachers (x-axis) and graduation rate (y-axis). Plot (c) shows the relationship between the number of instructional aides and paraprofessionals (x-axis) and graduation rate (y-axis). Plot (d) shows the relationship between the number of counselors (x-axis) and graduation rate (y-axis). Plot (e) shows the relationship between teacher salary (x-axis) and graduation rate (y-axis).

2.1 Common Core of Data

The Common Core of Data (CCD) is the United States Department of Education’s database on all public elementary and secondary schools and school districts (2025b). The variables of interest from this dataset are the number of students enrolled and whether or not a school is eligible for participation in either Target Assistance program or school-wide program authorized by Title I of Public Law 103-382.

Student enrollment reports the number of students enrolled in the school. Figure 1 (a) shows a logarithmic relationship, so a log transformation should be applied, but I will use a square root transformation because of the large number of zeros. A school is Title I eligible if it is in a school district for children from low-income families (2025c). Figure 1 (b) shows that schools that are not Title I eligible have a higher median graduation rate and is less varied, whereas schools that are Title I eligible are much more varied with a larger number of outliers. I picked these variables because small schools have been linked to higher graduation rates, and students from low-income families tend to have higher dropout rates (“High School Graduation,” n.d.).

2.2 The Civil Rights Data Collection

The Civil Rights Data Collection (CRDC) comes from the United States Department of Education’s Office for Civil Rights. This data is collection in order to ensure schools give students equal access to educational opportunities (2025e). Data are collected from public schools and districts, justice facilities, charter schools, alternative schools, and special education schools (2025e).

The key variables of interest from this dataset are the number of students enrolled in the International Baccalaureate Diploma Program, the number of students enrolled in the gifted and talented programs, the number of students enrolled in at least one AP course, the number of students that took one or more AP exams, the number of students participating in the SAT or ACT, the number of suspensions, the number of full-time teachers, number of first-year teachers, number of full-time equivalent instructional aides or paraprofessionals, number of full-time equivalent school counselors, and teacher salary. The number of students enrolled in the International Baccalaureate Diploma Program, the number of students enrolled in the gifted and talented programs, and the number of students enrolled in at least one AP course were transformed to the proportion of students enrolled in each respective program because that is more informative. Figure 2 (a) shows a logarithmic relationship between these programs and graduation rate. Figure 2 (b) and (c), Figure 1 (c), and all plots in Figure 3 also show logarithmic relationships between the selected variables and graduation rate, so a square root transformation is appropriate.

I picked these variables because students participating in higher level courses and/or participating in college-preparatory exams would be most interested in graduating and going to college. Suspensions would make it more difficult for students to graduate. Increased support

from experienced teachers and other aides can motivate students to do well in school. Higher teacher salary will motivate teachers to put more effort in helping their students.

2.3 *EDFacts*

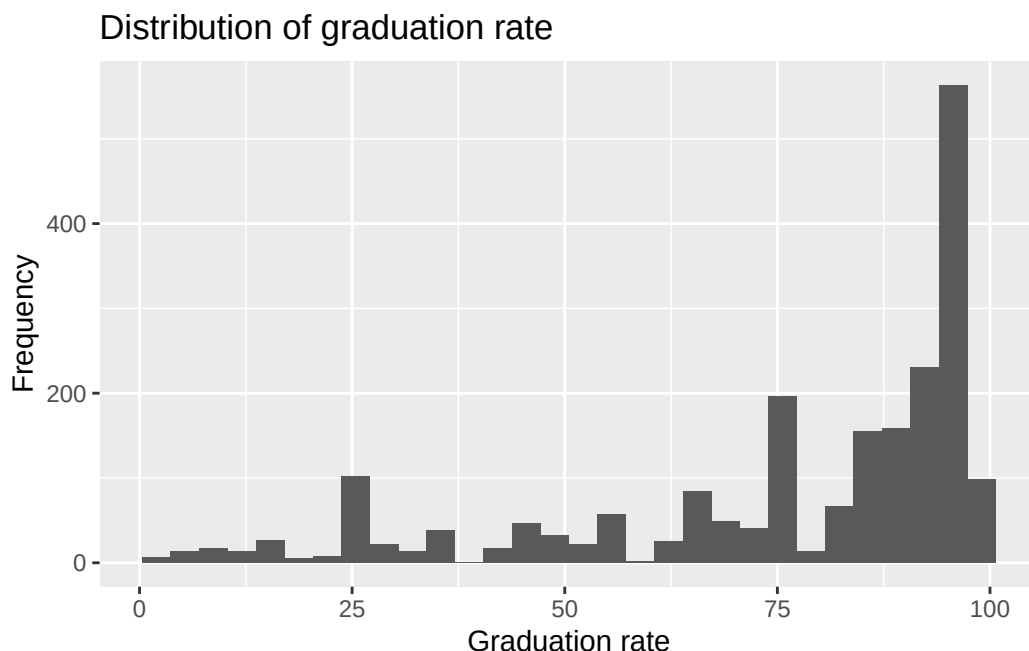


Figure 4: Histogram of graduation rate showing a left skewed distribution.

High school graduation rate comes from the *EDFacts* dataset. Figure 4 shows a left-skewed distribution in graduation rate.

EDFacts collects and analyzes data from pre-kindergarten through twelfth grade supplied by state education agencies with other data within the U.S. Department of Education (2025d).

2.4 Alternative Data Sources

A good alternate data source would be at the student level rather than the school level. This can be difficult to find, though, because the students are not legal adults and due to Family Educational Rights and Privacy Act (FERPA) which protects the distribution of educational records without the consent of parents or the student if they are over 18 years old. Information on absenteeism, family income level, extracurriculars of the students, involvement of parents and teachers, and reading level would also help in predicting graduation rates.

3 Methods

3.1 Base Model

In an attempt to answer this research question, I will first fit a multiple linear regression model using all variables and no transformations. Let Y be the 2125×1 vector where each entry represents the graduation rate at each school. Let X be the 2125×13 matrix where each column is a predictor. Let X_1 be the vector representing the number of students enrolled, X_2 be the vector equaling 1 if a school is Title I eligible and 0 if it's not, X_3 be the vector representing the proportion of students enrolled in the International Baccalaureate Diploma Program, X_4 be the vector representing the proportion of students enrolled in the AP program, X_5 be the vector representing the proportion of students enrolled in the gifted and talented program, X_6 be the vector representing the number of students that took one or more AP exams, X_7 be the vector representing the number of students participating in the SAT or ACT, X_8 be the vector representing the number of suspensions, X_9 be the vector representing the number of full-time teachers, X_{10} be the vector representing the number of first-year teachers, X_{11} be the vector representing the number of full-time equivalent instructional aides or paraprofessionals, X_{12} be the vector representing the number of full-time equivalent school counselors, and X_{13} be the vector representing teacher salary.

The model can be written as

$$Y = X\beta + \varepsilon$$

where β is the vector of coefficients for each predictor: β_0 represents the mean of the probability distribution of the graduation rate when all predictors are 0, β_1 represents the average change in graduation rate for an additional student enrolled, holding all other variables constant, β_2 represents the average change in graduation rate for a Title I school, holding all other predictors constant, β_3 represents the average change in graduation rate for an additional student enrolled in the IB program, holding all other variables constant, β_4 represents the average change in graduation rate for an additional student enrolled in the AP program, holding all other variables constant, β_5 represents the average change in graduation rate for an additional student enrolled in the gifted and talented program, holding all other variables constant, β_6 represents the average change in graduation rate for an additional student that took one or more AP exams, holding all other variables constant, β_7 represents the average change in graduation rate for an additional student participating in the SAT or ACT, holding all other variables constant, β_8 represents the average change in graduation rate for an additional suspension instance, holding all other variables constant, β_9 represents the average change in graduation rate for an additional full-time teacher, holding all other variables constant, β_{10} represents the average change in graduation rate for an additional first-year teacher, holding all other variables constant, β_{11} represents the average change in graduation rate for an additional full-time equivalent instructional aide or paraprofessional, holding all other variables constant, β_{12} represents the average change in graduation rate for an additional full-time equivalent school counselor, holding all other variables constant, β_{13} represents the average

change in graduation rate for a one USD increase in teacher salary, holding all other variables constant. ε is the vector containing the error.

3.2 Transformed Model

The next model used is the model where each predictor except Title I eligibility is transformed using the square root. While a log transformation would have been ideal, there are many zeros in the dataset which is not an acceptable input in a logarithmic function. A square root transformation is similar to a logarithmic one and allows zeros in its domain which makes it a viable substitution.

The model can be written as:

$$Y = \sqrt{X}\beta + \varepsilon$$

where each variable is equivalent to those in the base model.

3.3 Least Absolute Shrinkage and Selection Operator (LASSO) Regression Model

The last model created is one using LASSO regression to perform variable selection. LASSO is a shrinkage or regularization regression method that has the ability to “shrink” estimated coefficients to zero, carrying out variable selection. Alongside this, it reduces the variance of parameter estimates. LASSO aims to minimize the equation:

$$\sum_{i=1}^n (Y_i - Z_i^T \beta)^2 + \lambda \sum_{j=1}^{p-1} |\beta_j|$$

where $Z_i = \sqrt{X_i}$, $\lambda \geq 0$ is a tuning parameter that controls the level of regularization, n represents the number of observations, and p is the number of predictors. Essentially, we are minimizing the error sum of squares (SSE) with a penalty term added to it.

When picking the best lambda value, `cv.glmnet()` uses cross-validation to pick the best lambda that produces a mean squared error within one standard error of the mean squared error of the true best lambda (Friedman, Hastie, and Tibshirani 2010a). This is because the true best lambda may overfit the model by setting many estimated coefficients to zero. The user is able to pick any lambda used by the function based on their own analysis. For ease, I will use the default best lambda.

3.4 Model Selection

To determine which model is the best, I will first split the data into a 70% training, 15% testing, and 15% validation set. Then, I will train the models on the training set and find the mean squared prediction error using the validation set. This is defined as

$$E[(Y^* - \hat{Y}^*)^2]$$

where Y^* is the true graduation rate and $\hat{Y}^*$ is the predicted graduation rate using the model. This is the expected squared difference between the prediction and the true value over the validation data population. The one that gives the best mean squared error will be the best model. Finally, I will use the testing set to estimate the performance, such as error rates, for the final model.

3.5 Assumptions of Linear Regression

The assumptions of using linear regression for inference are:

1. The variables used accurately reflect the quantities of interest, the model should include all predictors, and the model should be applicable for the research question.
2. The sample data is representative of the population of interest.
3. The mean function must be correct.
4. Errors must be independent.
5. The variance of the errors must be equal.
6. Errors must be normal.

Since the goal of this analysis is point prediction, the most important assumption that should be met is linearity. If my goal was to create prediction intervals, I would need to meet the normality of errors assumption. After the transformations, there appears to be a linear relationship between the predictors and graduation rate.

3.6 Limitations of Analysis

One limitation of this analysis is that it is at the school level rather than the student level. While there are many reasons as to why this is the case, it does limit the analysis, and predictions can only be made for schools. An analysis at the student level, though, would change the problem from linear regression to logistic regression since the response would be whether the student graduated or not. Another limitation is that interaction effects are not included in any models. This is because it reduces the interpretability even though it could improve performance. In a future analysis, researching which variables have a theoretical interaction with each other would build upon this. Additionally, there may be models better suited for this research question that I have not learned yet.

3.7 Software

I use the `lm()` function from the R programming language (R Core Team 2025) to create the base model and the transformed model. I use the `cv.glmnet()` function from the `glmnet` package (Friedman, Hastie, and Tibshirani 2010b) to use LASSO regression to select the most important variables.

4 Results

The final model chosen is the LASSO regression model. The estimated intercept term is 57, but since it is not possible for all predictors to be zero, this has little meaning. The estimated slope parameter for enrollment is 0.867 which represents the average increase in graduation rate for an additional square root of a student, holding all other predictors constant. Title I eligibility was removed in the process, resulting in an estimated parameter of 0. The estimated slope parameter for the number of suspensions is -0.087 which represents the average decrease in graduation rate for an additional square root of a suspension instance, holding all other predictors constant. The estimated slope parameter for the proportion of students in the IB program is 9.447 which represents the average increase in graduation rate for an additional square root of a unit increase in the proportion of students in the IB program, holding all other predictors constant. The estimated slope parameter for the proportion of students in the gifted and talented program is -2.677 represents the average decrease in graduation rate for an additional square root of a unit increase in the proportion of students in the gifted and talented program, holding all other predictors constant. The estimated slope parameter for the proportion of students in the AP program is 51.519 which represents the average increase in graduation rate for an additional square root of a unit increase in the proportion of students in the AP program, holding all other predictors constant. The estimated slope parameter for the proportion of students taking one or more AP exams is -17.214 which represents the average decrease in graduation rate for an additional square root of a unit increase in the proportion of students taking one or more AP exams, holding all other predictors constant. The estimated slope parameter for the proportion of students participating in the SAT or ACT is 22.371 which represents the average increase in graduation rate for an additional square root of a unit increase in the proportion of students participating in the SAT or ACT, holding all other predictors constant. The estimated slope parameter for the number of full time equivalent teachers is -3.15 which represents the average decrease in graduation rate for an additional square root of a unit increase in the number of full time equivalent teachers, holding all other predictors constant. The estimated slope parameter for the number of first year teachers is 1.552 which represents the average increase in graduation rate for an additional square root of a unit increase in the number of first year teachers, holding all other predictors constant. The estimated slope parameter for the number of instructional aids is -0.907 which represents the average decrease in graduation rate for an additional square root of a unit increase in the number of instructional aides, holding all other predictors constant. The estimated slope

parameter for the number of counselors is -0.707 which represents the average decrease in graduation rate for an additional square root of a unit increase in the number of counselors, holding all other predictors constant. Teacher salary was also removed by the LASSO model, resulting in an estimated parameter of 0.

Since LASSO regularization removed Title I eligibility and teacher salary from the model, it believed it was not as important in predicting graduation rate or may already be explained by some other variables. This may be because Title I eligibility is binary, and including It's surprising that many estimated parameters are negative when one would assume that, for example, having more students taking an AP exam would increase graduation rates rather than decrease. There may be additional factors at play behind the scene, especially since this is at the school level.

4.1 Estimating Performance

Table 1 presents the mean squared prediction error for each model created and the final model, with the base model having the highest MSPE and the LASSO model having the lowest MSPE, and the MSPE on the test set is comparable to that of the validation set for the same model.

Table 1: This table shows the mean squared prediction errors on the validation set for each created model and for the chosen model on the testing set.

Base Model MSPE (Validation)	Transformed Model MSPE (Validation)	LASSO Model MSPE (Validation)	Final Model MSPE (Test)
378.232	347.964	337.728	338.443

Clearly explain what the model results show and connect your results to the central research question(s) and their broader context

5 Discussion

The aim of this paper was to answer the following research question: Can college readiness, staff demographics, and school demographics predict graduation rates in California high schools? I investigated this using data from the Common Core of Data, The Civil Rights Data Collection, and *EDFacts*. Then, I created three models. The first was a multiple regression model with graduation rate as the response and the number of students enrolled, Title I eligibility, the proportion of students in the IB program, the proportion of students in the AP program, the proportion of students in the gifted and talented program, the proportion of students that took one or more AP exams, the proportion of students taking the SAT or ACT, the number of suspensions, the number of full-time teachers, the number of full-time first-year teachers,

the number of instructional aides and paraprofessionals, the number of counselors, and teacher salary as predictors. The second model transformed each predictor except Title I eligibility using a square root transformation. The last model was made using LASSO on the previous model to select the best predictors. The model that predicted the best out of the three was the LASSO model:

$$Y = \begin{bmatrix} | & | & | & | & | & | & | & | & | & | & | & | & | \\ 1 & X_1 & X_2 & X_3 & X_4 & X_5 & X_6 & X_7 & X_8 & X_9 & X_{10} & X_{11} & X_{12} & X_{13} \\ | & | & | & | & | & | & | & | & | & | & | & | & | & | \end{bmatrix} \begin{bmatrix} 57 \\ 0.867 \\ 0 \\ -0.087 \\ 9.447 \\ -2.677 \\ 51.519 \\ -17.214 \\ 22.371 \\ -3.15 \\ 1.552 \\ -0.907 \\ -0.707 \\ 0 \end{bmatrix}$$

The predictors with a parameter estimate of zero, Title I eligibility and teacher salary, were removed by LASSO regularization. This model has a test mean squared prediction error of 338.443. This means that college readiness, staff demographics, and school demographics can predict graduation rates, but Title I eligibility and teacher salary were not influential enough to improve predictions.

5.1 Potential Weaknesses

One weakness of this study is the lack of interaction terms. For example, knowing if a school is Title I eligible or not may impact the proportion of students who take an AP exam or SAT/ACT because one needs to pay to take those tests. Including them could explain more variability and increase the predictive power of the final model.

Another weakness is that there are more factors to graduation rate than the predictors I chose. Many of these factors, like family income level, if the student is first generation, and student responsibilities, are at the student level and not generally available or quantifiable. Graduation rates can also be affected by sudden changes, such as a death in the family, that may cause a decrease in performance or even cause the student to drop-out all together.

5.2 Possible Extensions

An extension of this study would be to put out a survey at the student level to find more information on the students' motivations and other factors mentioned above. Another extension would be to answer an inference-based research question, such as using an F-test to determine if any of the chosen predictors do not have a relationship with graduation rate. One could also go beyond and predict success in college and beyond using a different set of predictors, which is useful for college admissions.

6 Questions

1. should i be having this many graphs?? does figure 2 look ok only 4-5, maybe if 0 remove it as a proportion
2. do i need to explain lasso more, change x to z and explain how lambda is calculated
3. do i need to add more info for describing the dataset
4. should i also put information on mse of training? should i put other metrics besides mspe?
5. is my interpretation of the estimated parameters ok?
6. is the way i wrote the final model ok

References

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